

Proof. Theorem (Bernoulli's Inequality). Let $x \in \mathbb{R}$ satisfy $x \geq -1$ and let $n \in \mathbb{N}_0$. Then

$$(1+x)^n \geq 1+nx.$$

1. **Degenerate cases.**

- **$n = 0$.** $(1+x)^0 = 1$ and $1+0 \cdot x = 1$; equality holds.
- **$x = -1$.** For $n \geq 1$, $(1+x)^n = 0$ and $1+nx = 1-n \leq 0$; hence $0 \geq 1-n$. Equality occurs only when $n = 1$ (already covered by the case $n = 1$).
- **$x = 0$.** $(1+0)^n = 1$ and $1+n \cdot 0 = 1$; equality holds for every n .

Thus the theorem is true in all degenerate situations.

2. **Induction on the exponent.**

(a) **Base step** ($n = 1$).

$$(1+x)^1 = 1+x = 1+1 \cdot x,$$

so the inequality holds with equality.

(b) **Inductive hypothesis.** Assume that for a fixed integer $k \geq 1$

$$\boxed{(1+x)^k \geq 1+kx} \quad \text{for every } x \geq -1. \quad (\text{H}_k)$$

(c) **Inductive step.** Since $x \geq -1$ we have $1+x \geq 0$. Multiplying both sides of (H_k) by this non-negative factor preserves the inequality:

$$\begin{aligned} (1+x)^{k+1} &= (1+x)(1+x)^k \\ &\geq (1+x)(1+kx). \end{aligned} \quad (1)$$

Expanding the product gives the identity

$$(1+x)(1+kx) = 1 + (k+1)x + kx^2. \quad (2)$$

Because $k \geq 1$ and $x^2 \geq 0$, the term kx^2 is non-negative; discarding it yields

$$1 + (k+1)x + kx^2 \geq 1 + (k+1)x. \quad (3)$$

Combining (1), (2) and (3) we obtain

$$(1+x)^{k+1} \geq 1 + (k+1)x.$$

Thus the statement holds for $n = k+1$.

By the principle of mathematical induction, (H_n) is true for every integer $n \geq 0$; i.e.

$$(1+x)^n \geq 1+nx \quad \forall x \geq -1, n \in \mathbb{N}_0. \quad (*)$$

3. **Equality conditions.** Equality can arise only in the steps where we did not introduce a strict inequality:

- In the base cases $n = 0$ and $n = 1$ we have exact equality for all admissible x .
- In the inductive step the only possible loss of equality is the removal of the non-negative term kx^2 in (3). Equality there forces $kx^2 = 0$; since $k \geq 1$ this yields $x^2 = 0$, i.e. $x = 0$.

Hence equality holds *iff* one of the following occurs:

- (a) $n = 0$ (any $x \geq -1$);
- (b) $n = 1$ (any $x \geq -1$);
- (c) $x = 0$ (any $n \geq 0$).

No other combination gives equality, because for $n \geq 2$ and $x \neq 0$ the term $kx^2 > 0$ appears at the first inductive step and yields a strict inequality that persists for larger n .

4. **Conclusion.** For every real number $x \geq -1$ and every integer $n \geq 0$,

$$\boxed{(1+x)^n \geq 1+nx},$$

with equality precisely in the cases listed above. $\square$

$\square$